

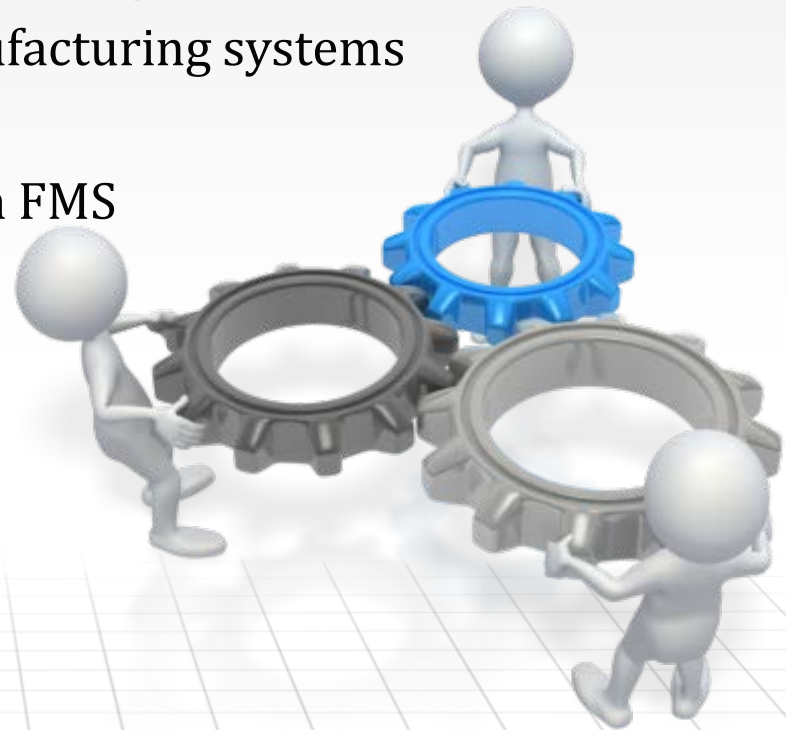


LECTURE 7

FLEXIBLE MANUFACTURING SYSTEMS

Contents

1. Objectives
2. Review
3. Flexible manufacturing system (FMS)
4. Flexibility, productivity and costs
5. Types of flexibility in manufacturing
6. Classification of flexible manufacturing systems
7. FMS components
8. Methods for the analysis of an FMS
9. Conclusion



Objectives

1. Defining concepts of flexibility, productivity and cost.
2. Knowing the types of flexibility.
3. Classification of flexible manufacturing systems (FMS).
4. Knowing the components of FMS and the relationships between them.
5. Presenting the analysis methods of an FMS.



Review

1. Group technology
2. Parts classification and coding
3. Production flow analysis



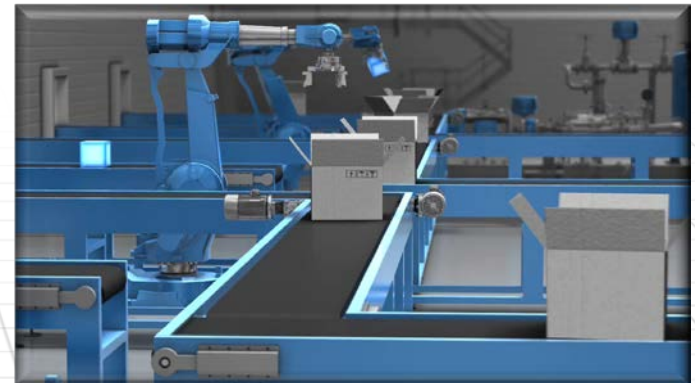


Flexible manufacturing system

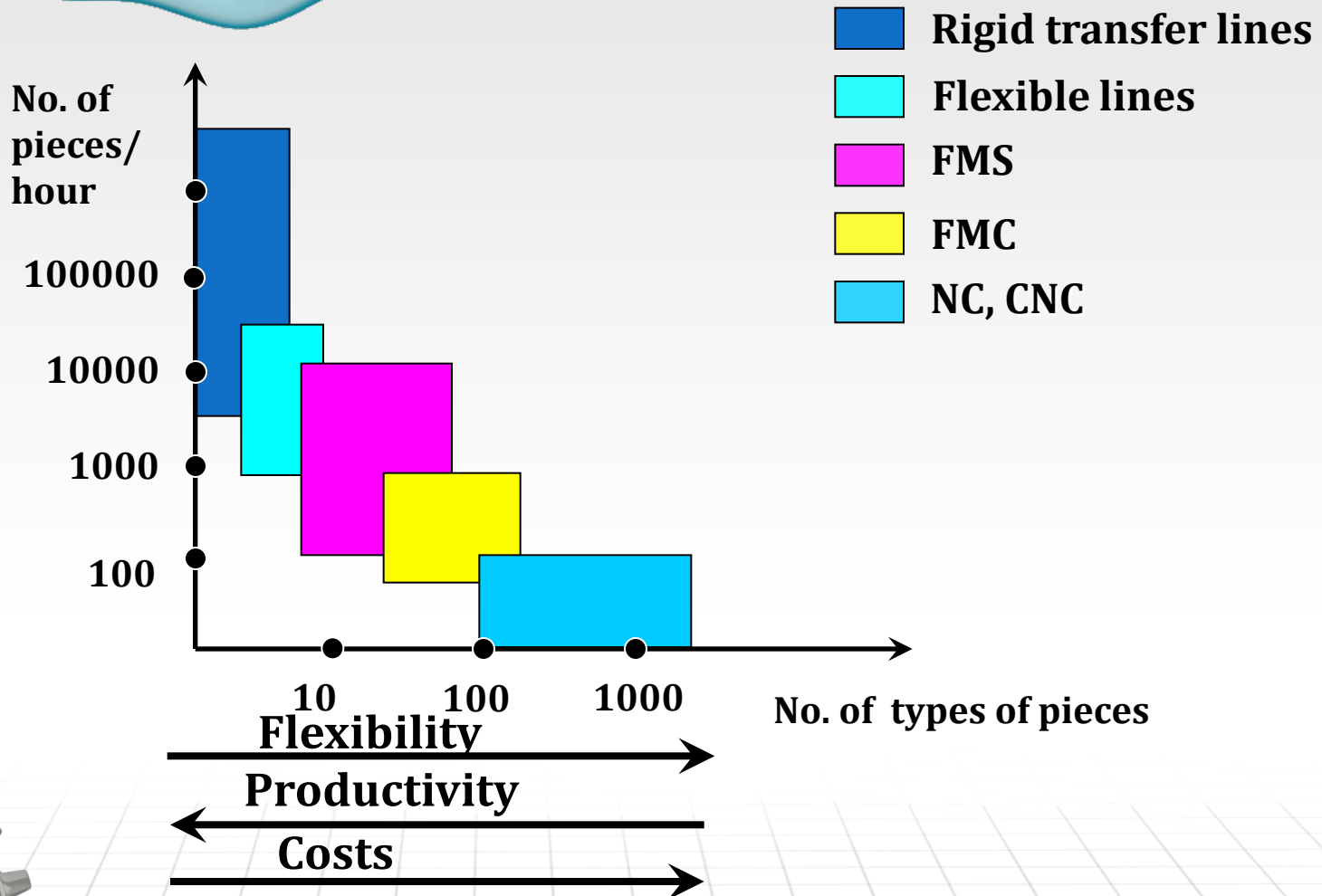
A flexible manufacturing system is a highly automated GT machine cell, consisting of a group of processing workstations (usually CNC machine tools), interconnected by an automated material handling and storage system, and controlled by a distributed computer system.

In order to be flexible a manufacturing system must possess:

- the ability to **identify** and **distinguish** among the different part or product styles processed by the system,
- quick **changeover** of operating **instructions**,
- quick **changeover** of physical **setup**



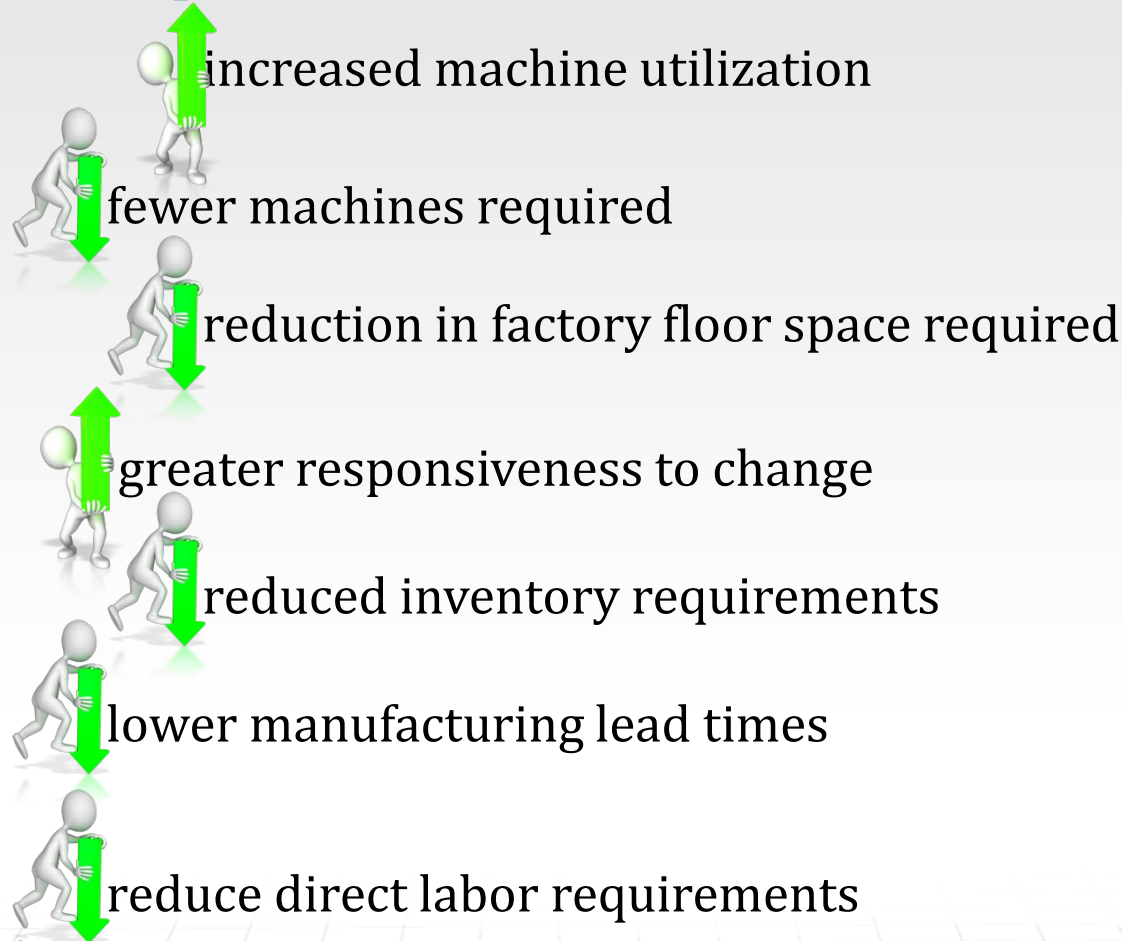
Flexibility Productivity Costs



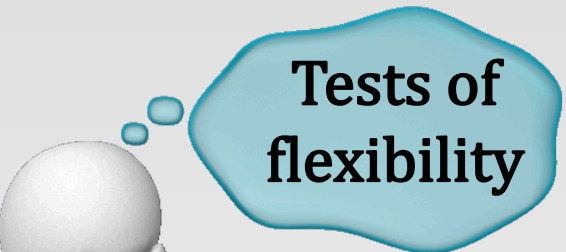
FMS – Flexible Manufacturing System; CNC – Computer Numerical Control
CIM – Computer Integrated Manufacturing; FMS – Flexible Manufacturing System; FMC – Flexible Manufacturing Cell; NC – Numerical Control;



The benefits of an FMS



There is the opportunity for unattended production.



Tests of flexibility



Can the system process different part styles in a non batch mode?

Part variety test



Can the system accept changes in production scheduling and changes in either part mix or production quantities?

Schedule change test



Can the system recover gracefully from equipment malfunction and breakdowns, so that production is not completely disrupted?

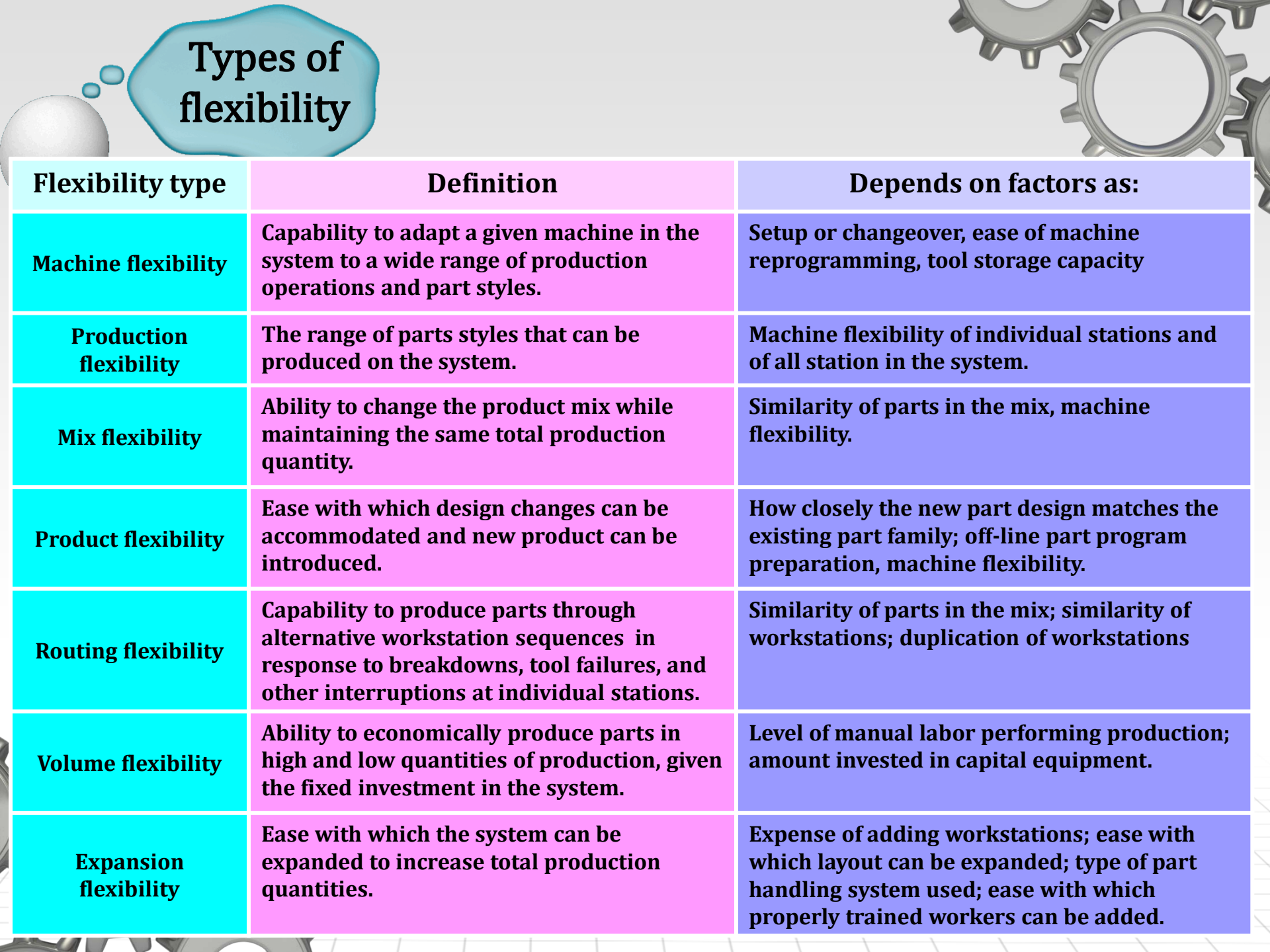
Error recovery test



Can new part designs be introduced into the existing product mix with relative ease?

New part test





Types of flexibility

Flexibility type	Definition	Depends on factors as:
Machine flexibility	Capability to adapt a given machine in the system to a wide range of production operations and part styles.	Setup or changeover, ease of machine reprogramming, tool storage capacity
Production flexibility	The range of parts styles that can be produced on the system.	Machine flexibility of individual stations and of all station in the system.
Mix flexibility	Ability to change the product mix while maintaining the same total production quantity.	Similarity of parts in the mix, machine flexibility.
Product flexibility	Ease with which design changes can be accommodated and new product can be introduced.	How closely the new part design matches the existing part family; off-line part program preparation, machine flexibility.
Routing flexibility	Capability to produce parts through alternative workstation sequences in response to breakdowns, tool failures, and other interruptions at individual stations.	Similarity of parts in the mix; similarity of workstations; duplication of workstations
Volume flexibility	Ability to economically produce parts in high and low quantities of production, given the fixed investment in the system.	Level of manual labor performing production; amount invested in capital equipment.
Expansion flexibility	Ease with which the system can be expanded to increase total production quantities.	Expense of adding workstations; ease with which layout can be expanded; type of part handling system used; ease with which properly trained workers can be added.



Self-assessment test

1. An FMS needs less time than a classical manufacturing system to adapt to a new product.

T

F

2. Increased flexibility leads to higher costs.

T

F

3. Stock levels decreases by using an FMS.

T

F

4. In an FMS occurrence of errors should not stop completely the production.

T

F



FMS classification

- ☐ By the number of machines:
 - single machine cell
 - flexible manufacturing cell
 - flexible manufacturing system

- ☐ By the level of flexibility:
 - dedicated FMS
 - random-order FMS

A **single machine cell** consists of one CNC machining center combined with a parts storage system for unattended operation.

A **flexible manufacturing cell** consists of two or three processing workstations plus a handling system.

A **flexible manufacturing system** has four or more processing workstations connected mechanically by a common part handling system and electronically by a distributed computer system.

A **dedicated FMS** is designed to produce a limited variety of part styles, and the complete universe of parts to be made on the system is known in advance.

A **random-order FMS** is more appropriate when the part family is large, there are substantial variation in part configurations, there will be new part design introduced, and the production scheduling is subject to change from day-to-day.

FMS components

Basic components of an FMS:

- workstations
 - load/unload stations
 - machining stations
 - other processing stations
 - assembly stations
 - other stations and equipment
- material handling and storage system
- computer control system
- human resources





FMS planning and design



FMS planning considers the parts that will be produced and include:

- part family considerations
- processing requirements
- physical characteristics of the workparts
- production volume

Important factors that must be specified in FMS design:

- 
- types of workstations
 - variations in process routings and FMS layout
 - material handling system
 - work-in-process and storage capacity
 - tooling
 - pallet fixtures



Quantitative analysis of FMS

❑ FMS analysis techniques can be classified as follows:

- deterministic models
- queueing models
- discrete event simulation
- other approaches including heuristics

- *Deterministic models* can be used to obtain starting estimates of system performance.
- *Queueing models* can be used to describe some of the dynamics not accounted for in deterministic approaches.
- *Discrete event simulation* offers the most accurate method for modeling the specific aspects of a given FMS. (ex. ARENA software)

➤ Other techniques:

- mathematical programming
- various heuristics approaches



Self-assessment test

1. A flexible manufacturing cell has minimum four machines and a materials handling system.

T

F

2. A dedicated FMS is less useful than an universal one.

T

F

3. Human resource is not a component of an FMS.

T

F

4. Designing an FMS requires information about the physical characteristics of the products.

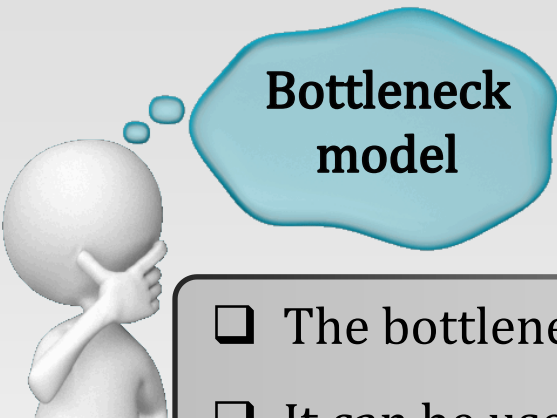
T

F

5. Discrete event simulation offers the most accurate method for modeling an FMS.

T

F



Bottleneck model

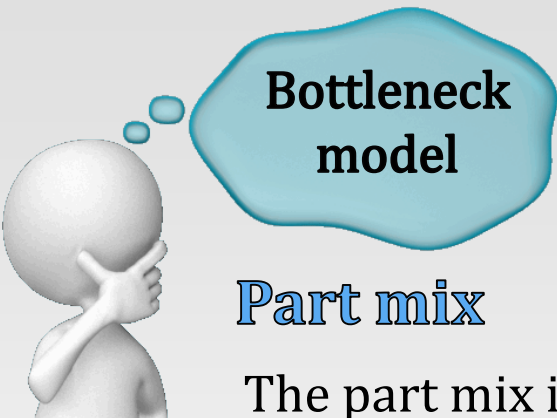
- ❑ The bottleneck model is a deterministic model developed by Solberg.
- ❑ It can be used to provide starting estimates of FMS design parameters such as production rate and number of workstations.

Terminology used in bottleneck model:

- Part mix
- Workstations and servers
- Process routing
- Work handling system
- Transport time
- Operation frequency



The term bottleneck refers to the fact that the output of the production system has an upper limit, given that the product mix flowing through the system is fixed.



Bottleneck model

Part mix

The part mix is defined by the equation:

$$\sum_{j=1}^P p_j = 1.0$$

where:

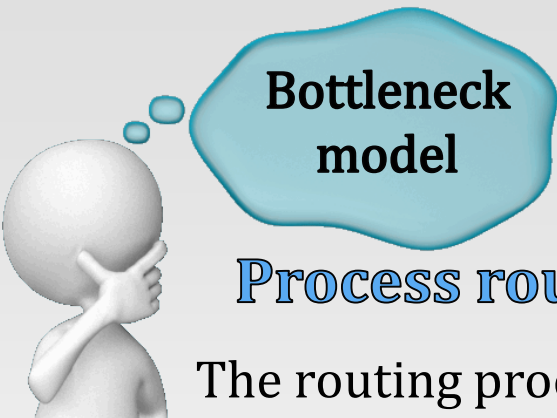
p_j – the fraction of the total system output that is of style j

P – total number of different part styles made in the FMS

Workstations and servers

In the terminology of the bottleneck model, each workstation may have more than one station or server (machine capable to perform the same operations).

s_i – the number of servers at workstation i
 $i=1..n$, where n – number of workstations



Bottleneck model

Process routing

The routing process defines for each part or product:

- the sequence of operations,
- the workstations at which it is produced,
- the associated processing times.

t_{ijk} - the processing time – is the total time that a production unit occupies a given workstation server, not counting any waiting time at the station.

i – number of station

j – number of part or product

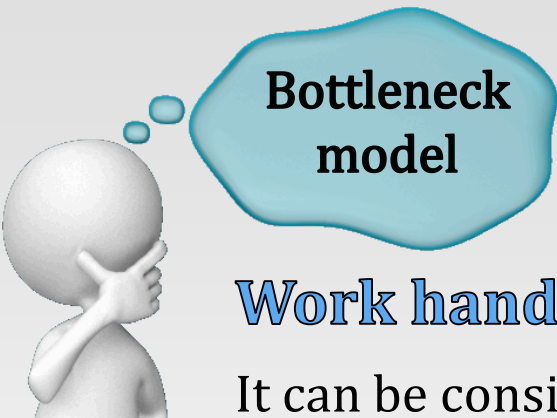
k – the sequence of operations in the process routing



Example 1

The fourth operation in the process plan for part A is performed on machine 2 and takes 8.5 min

$$\Rightarrow t_{2A4} = 8.5 \text{ min}$$



Bottleneck model

Work handling system

It can be considered to be a special case of a workstation (let it be station $n+1$).

Let s_{n+1} - the number of carriers in the system (conveyor carts, AGVs, etc)

Transport time

Let t_{n+1} - the mean transport time required to move a part from one workstation to the next station in the process routing.

Operation frequency

It is the expected number of times a given operation in the process routing is performed for each work unit

Let f_{ijk} - the operation frequency for operation k in part j at station i .



Bottleneck model

Workload of stations



- ❑ The average workload for a given station is the mean total time spent at the station per part:

$$WL_i = \sum_j \sum_k \left(t_{ijk} \times f_{ijk} \times p_j \right)$$

where:

WL_i – average workload for station i (min)

t_{ijk} – processing time for operation k in process plan j at station i

f_{ijk} – operation frequency for operation k in part j at station i

p_j – part mix fraction for part j

- ❑ The workload for the handling system is:

$$WL_{n+1} = n_t \times t_{n+1}$$

$$n_t = \sum_i \sum_j \sum_k \left(f_{ijk} \times p_j \right) - 1$$

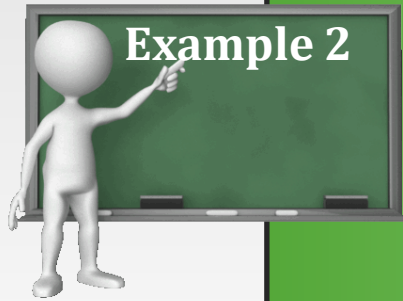
where:

WL_{n+1} – workload of the handling system (min)

n_t – mean number of transports

t_{n+1} – mean transport time per move (min)

The workload for the handling system



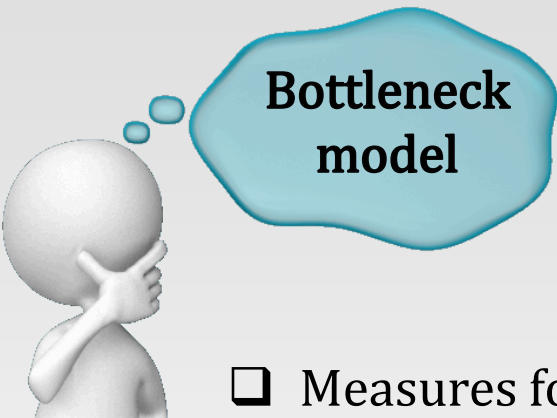
Consider a manufacturing system with two stations: (1) a load/unload station and (2) a machine station. There is just one part processed through the production system, part A, so the part mix fraction $p_A = 1.0$. The frequency of all operations is $f_{iAk} = 1.0$. The parts are loaded at station 1, routed to station 2 for machining, and then sent back to station 1 for unloading (three operations in the routing). Determine the average number of transports.

Solution

$$n_t = \sum_i \sum_j \sum_k (f_{ijk} \times p_j) - 1 = 1 \times (1.0) + 1 \times (1.0) + 1 \times (1.0) - 1 = 3 - 1 = 2$$

The process routing is: (1) $\rightarrow$ (2) $\rightarrow$ (1)

Counting the number of arrows gives us the number of transports: $n_t = 2$



Bottleneck model

System performance measures

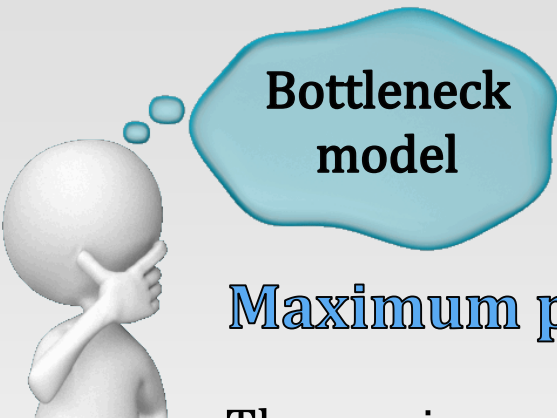


❑ Measures for assessing the performance of an FMS:

- production rate of all parts,
- production rate of each part style,
- utilization of the different workstations,
- number of busy servers at each workstation.



- These measures can be calculated under the assumption that the FMS is producing at its maximum possible rate.
- The maximum possible rate is constrained by the bottleneck station in the system which is the station with the highest workload per server. (WL_i/s_i)



Bottleneck model

Maximum production rate of all parts

The maximum production rate of all parts is the maximum production rate of the bottleneck station:

$$R_p^* = \frac{s^*}{WL^*}$$

where:

R_p^* - maximum production rate of all part styles (pc/min)

s^* - number of servers at the bottleneck station

WL^* - workload at the bottleneck station (min/pc)

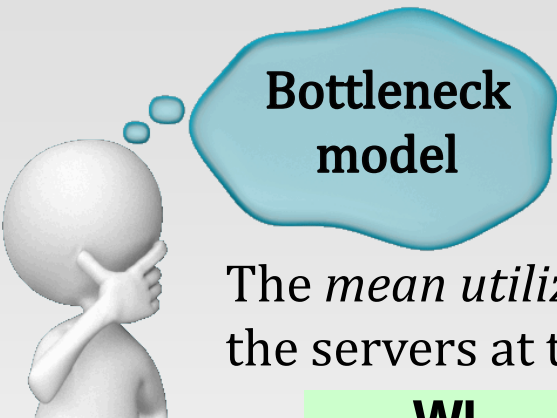
Production rate of part style

$$R_{pj}^* = p_j \times R_p^* = p_j \times \frac{s^*}{WL^*}$$

where:

R_{pj}^* - maximum production rate of part style j (pc/min)

p_j - part mix fraction for part style j



Bottleneck model

Utilization of workstation

The *mean utilization* of each workstation is the proportion of time that the servers at the station are working and not idle:

$$U_i = \frac{WL_i}{s_i} \times R_p^* = \frac{WL_i}{s_i} \times \frac{s^*}{WL^*}$$

$$\bar{U} = \frac{\sum_{i=1}^{n+1} U_i}{n+1}$$

$$\bar{U}_s = \frac{\sum_{i=1}^n s_i \times U_i}{\sum_{i=1}^n s_i}$$

where: U_i – utilization of station i

s_i – number of servers at station i

WL_i – workload of station i (min/pc)

R_p^* – overall production rate (pc/min)

$\bar{U}$ – unweighted average of the workstation utilizations

$\bar{U}_s$ – overall FMS utilization

Number of busy servers

$$BS_i = WL_i \times R_p^* = WL_i \times \frac{s^*}{WL^*}$$

where: BS_i number of busy servers on average at station i

WL_i – workload at station i

- A more useful measure of overall FMS utilization can be obtained using a weighted average.
- The weighting is based on the number of servers at each station for the n regular stations in the system.
- The transport system is omitted from the average.

System performance measures

Example 3

A flexible manufacturing system consists of two machining workstations and a load/unload station. Station 1 is the load/unload station. Station 2 performs milling operations and consists of two servers (two identical CNC milling machines). Station 3 has one server that performs drilling (one CNC drill press). The stations are connected by a part handling system that has four work carriers. The mean transport time is 3.0 min. The FMS produces two parts, A and B. The part mix fractions and process routings for the two parts are presented below. The operation frequency $f_{ijk} = 1.0$ for all operations. Determine: (a) maximum production rate of the FMS, (b) corresponding production rates of each product, (c) utilization of each station, and (d) number of busy servers at each station.

Part j	Part mix p_j	Operation k	Description	Station i	Process time t_{ijk} (min)
A	0.4	1	Load	1	4
		2	Mill	2	30
		3	Drill	3	10
		4	Unload	1	2
B	0.6	1	Load	1	4
		2	Mill	2	40
		3	Drill	3	15
		4	Unload	1	2

System performance measures

Example 3

Solution

(a) To compute the FMS production rate, first it needs to compute workloads at each station in order to identify the bottleneck station:

$$WL_i = \sum_j \sum_k (t_{ijk} \times f_{ijk} \times p_j)$$

$$WL_1 = (4+2) \times 0.4 \times 1.0 + (4+2) \times 0.6 \times 1.0 = 6 \text{ min}$$

$$WL_2 = 30 \times 0.4 \times 1.0 + 40 \times 0.6 \times 1.0 = 36 \text{ min}$$

$$WL_3 = 10 \times 0.4 \times 1.0 + 15 \times 0.6 \times 1.0 = 13.0 \text{ min}$$

$$WL_4 = 3 \times 3.0 \times 0.4 \times 1.0 + 3 \times 3.0 \times 0.6 \times 1.0 = 9.0 \text{ min}$$

The station routing for both parts is the same (1) -> (2) -> (3) -> (1) => $n_t = 3$

The bottleneck station is identified by finding the largest WL_i/s_i ratio

$$\frac{WL_1}{s_1} = \frac{6.0}{1} = 6.0 \text{ min}$$

$$\frac{WL_2}{s_2} = \frac{36.0}{2} = 18.0 \text{ min}$$

$$\frac{WL_3}{s_3} = \frac{13.0}{1} = 13.0 \text{ min}$$

$$\frac{WL_4}{s_4} = \frac{9.0}{4} = 2.25 \text{ min}$$

$$R_p^* = \frac{s^*}{WL^*}$$

$$R_p^* = \frac{2}{36.0} = 0.05555 \text{ pc/min} = 3.333 \text{ pc/hr}$$

(b) To determine production rate of each product, multiply R_p^* by its respective part mix fraction:

$$R_{pA}^* = p_A \times R_p^* = 0.4 \times 3.333 = 1.333 \text{ pc/hr}$$

$$R_{pB}^* = p_B \times R_p^* = 0.6 \times 3.333 = 2.00 \text{ pc/hr}$$

System performance measures

Example 3

Solution

(continuation)

(c) The utilization of each station:

$$U_1 = \left(\frac{6.0}{1} \right) \times 0.05555 = 0.333 \quad (33.3\%)$$

$$U_2 = \left(\frac{36.0}{2} \right) \times 0.05555 = 1.00 \quad (100\%)$$

$$U_3 = \left(\frac{13.0}{1} \right) \times 0.05555 = 0.722 \quad (72.2\%)$$

$$U_4 = \left(\frac{9.0}{4} \right) \times 0.05555 = 0.125 \quad (12.5\%)$$

(d) Mean number of busy servers at each station:

$$BS_1 = WL_1 \times R_p^* = 6.0 \times 0.05555 = 0.333$$

$$BS_2 = WL_2 \times R_p^* = 36.0 \times 0.05555 = 2.0$$

$$BS_3 = WL_3 \times R_p^* = 13.0 \times 0.05555 = 0.722$$

$$BS_4 = WL_4 \times R_p^* = 9.0 \times 0.05555 = 0.50$$

$$U_i = \frac{WL_i}{s_i} \times R_p^* = \frac{WL_i}{s_i} \times \frac{s^*}{WL^*}$$

$$BS_i = WL_i \times R_p^* = WL_i \times \frac{s^*}{WL^*}$$

System performance measures

Example 3

Solution

(another approach)

Given that station 2 is the bottleneck, let us try to verify the maximum production rate of the FMS.

Processing times at station 2 are: $t_{2A2}=30$ min; $t_{2B2}=40$ min

Part mix fractions are: $p_A=0.4$; $p_B=0.6 \Rightarrow$ for every unit of B produced, there are $\left(\frac{0.4}{0.6}\right)=\frac{2}{3}$ units of part A

The corresponding time to process 1 unit of B and $\frac{2}{3}$ unit of A at station 1 is:

$$\frac{2}{3} \times 30 + 1 \times 40 = 20 + 40 = 60 \text{ min}$$

With two servers, the FMS can produce parts at the following maximum rate:

$$R_p^* = 2 \times \left(\frac{2}{3} + 1 \right) = 2 \times 1.6666 = 3.333 \text{ pc/hr}$$

System performance measures

Example 3

Solution

(another approach)

Given that the bottleneck station is working 100% utilization, let us try to determine the utilizations of the other stations:

At station 1, the time needed to load and unload the output of the two servers at station 2 is:

$$3.333 \times (4+2) = 20.00 \text{ min}$$

As a fraction of 60 min in an hour, this gives an utilization of:

$$U_1 = 0.333$$

At station 3 the processing time required to process the output of the two servers at station 2 is:

$$\frac{4}{3} \times 10 + 2 \times 15 = 43.333 \text{ min}$$

As a fraction of the 60 min, we have:

$$U_3 = 43.333 / 60 = 0.722$$

For the part handling system we have:

$$\frac{4}{3} \times 9.0 + 2 \times 9 = 30 \text{ min}$$

As a fraction of 60 min, this is 0.5; there are 4 carts =>

$$U_4 = 0.125$$



Bottleneck model

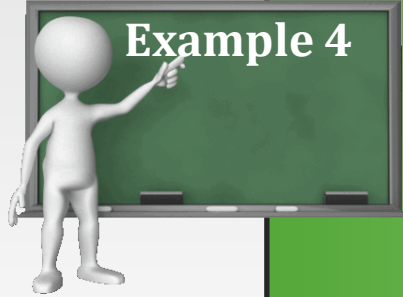
Sizing the FMS

- ❑ The bottleneck model can be used to calculate the number of servers required at each workstation to achieve a specified production rate.
- ❑ The starting information needed:
 - part mix
 - process routings
 - processing times
- ❑ The number of servers at each station i is determined by:

$$s_i = \text{minimum integer} \geq R_p \times WL_i$$

where: s_i – number of server at station i
 R_p – specified production rate of all parts to be produced
 WL_i – workload at station i

Sizing the FMS



Example 4

Suppose the part mix, process routings, and processing times for the family of parts to be machined on a proposed FMS are those given in Example 3. Determine how many servers at each station i will be required to achieve an annual production rate of 20,000 parts/yr. The FMS will operate 24 hr/day, 5 day/wk, 50 wk/yr. Anticipated availability of the system is 95%.

Solution:

The average hourly production rate is given by:

$$R_p = \frac{20000 \text{ pc / yr}}{0.95 \times (24 \times 5 \times 50) \text{ hr / yr}} = 3.5087 \text{ pc / hr} = 0.05848 \text{ pc / min.}$$

Using equation: $s_i = \text{minimum integer} \geq R_p \times WL_i$

$$s_1 = \text{minimum integer} \geq (0.05848 \times 6.0 = 0.3508) = 1 \text{ server}$$

$$s_2 = \text{minimum integer} \geq (0.05848 \times 36.0 = 2.1053) = 3 \text{ servers}$$

$$s_3 = \text{minimum integer} \geq (0.05848 \times 13.0 = 0.76024) = 1 \text{ server}$$

$$s_4 = \text{minimum integer} \geq (0.05848 \times 9.0 = 0.52632) = 1 \text{ server}$$

Sizing the FMS

Example 5

For the specified production rate in Example 4, determine: (a) the utilization of each station and (b) the maximum possible production rate if the utilization of the bottleneck station were increased to 100%.

Solution:

(a) The utilization at each workstation is determined as the calculated value of s_i divided by the resulting minimum integer $\geq s_i$:

$$U_1 = \frac{0.3508}{1} = 0.3508 \quad (35.08\%)$$

$$U_3 = \frac{0.76024}{1} = 0.76024 \quad (76.02\%)$$

$$U_2 = \frac{2.1053}{3} = 0.7017 \quad (70.17\%)$$

$$U_4 = \frac{0.52632}{1} = 0.52632 \quad (52.63\%)$$

The maximum value is at station 3 so this is the bottleneck station.

(b) The maximum production rate of the FMS is:

$$R_p^* = \frac{3.5087}{0.76024} = 4.6152 \text{ pc/hr} = 0.0769 \text{ pc/min.}$$

The corresponding utilization is: $U^* = U_3 = 0.0769 \times \frac{13}{1} = 1.0 \quad (100\%)$



Conclusion

- ☐ The flexibility of a manufacturing system depends on its ability to identify and discriminate between the various components as well as the ability to quickly adapt to a new product.
- ☐ Generally increased flexibility leads to lower productivity and increased costs.
- ☐ Today, to be competitive on the market the manufacturer must use manufacturing systems with high flexibility.



- ☐ Process planning
- ☐ Concurrent engineering
- ☐ Design for manufacturing and assembly
- ☐ Advanced manufacturing planning